

V S MONISH KUMAR

Software Engineer

vsmonishkumar24@gmail.com | 8688470533 | Chennai, India | [LinkedIn](#) | [GitHub](#)

TECHNICAL SKILLS

AI/ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), TensorFlow, Scikit-Learn, Natural Language Processing (NLP), Stable Diffusion, SAM (Segment Anything Model), U-Net

Dev & Tools: Python, Java, ETL, SQL, Rest API (Flask & Fast), Git, AWS, GCP, JIRA, Full Stack Development (Mobile & Web) Automation QA (Playwright) API Automation testing.

EXPERIENCE

Software Engineer | CLUSTREX DATA PRIVATE LIMITED

Jan 2025 – Present

- Developed AI-powered computer vision systems using Stable Diffusion, U-Net, and Segment Anything Model (SAM) for image generation and high-precision segmentation tasks.
- Built and fine-tuned a 37B parameter LLM-based resume parser, leveraging RAG pipelines and optimization techniques for improved accuracy and automation.
- Designed and implemented scalable REST APIs using Flask and FastAPI, integrating OTP-based authentication, and Twilio services.
- Developed a Google Chat-based bot system with:
 - Policy Assistant Bot using RAG architecture on Vertex AI, enabling intelligent retrieval from structured policy corpora.
 - Office Management Bot to track and manage employee leave, work-from-home (WFH), and attendance (punch data). Which generates the monthly summary that is used for salary generation.
- Developed cross-platform mobile applications using React Native and Android (Kotlin) with responsive UI and seamless backend API integration.
- Deployed and managed applications on AWS and GCP.
- Designed modular and scalable system architectures, focusing on reusability, maintainability, and performance optimization.
- Performed end-to-end testing on production applications, including manual UI/UX validation, data testing, and automation using Playwright; identified and reported bugs, enhancements, and issues using Git and JIRA.
- Implemented DevOps and security best practices, including monitoring, automated backups, database optimization, and secure code reviews.

Research Intern | SSN College of Engineering [Repository](#) [Certificate](#)

May 2024 - Jul 2024

- Built an IoT-based pedestrian detection system using YOLO and Arduino for real-time vehicle safety.

PROJECTS

Netflix TV Shows and Movies Analytics

[Repository](#)

CAT AND DOG IMAGE CLASSIFIER

[Repository](#)

IBM Cloud Watson Assistant For Facebook

[Repository](#)

Emotion Detection Using Computer Vision

[Repository](#)

EDUCATION

PRATHYUSHA ENGINEERING COLLEGE

Nov 2021 – Apr 2025 | Chennai

Bachelor of Engineering (B.E) – Computer Science and Engineering

CGPA: 9.08 / 10

CERTIFICATIONS

AWS Academy Cloud Foundations

[Certificate](#) | [Verify](#)

AWS Academy Machine Learning for Natural Language Processing

[Certificate](#) | [Verify](#)

Data Analytics

[Certificate](#) | [Verify](#)

Python for Data Science

[Certificate](#) | [Verify](#)